

June 2020

CS293B Course Project

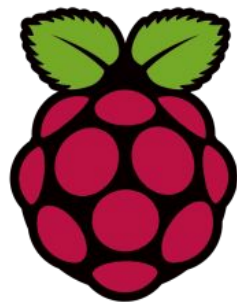
Deep Learning and Object Classification at the Edge

Project Plan

- Configure a system that uses a robot's camera feed and a trained deep learning model to perform object classification at the Edge.
 - Architecture similar to *Where's The Bear?*



IoT



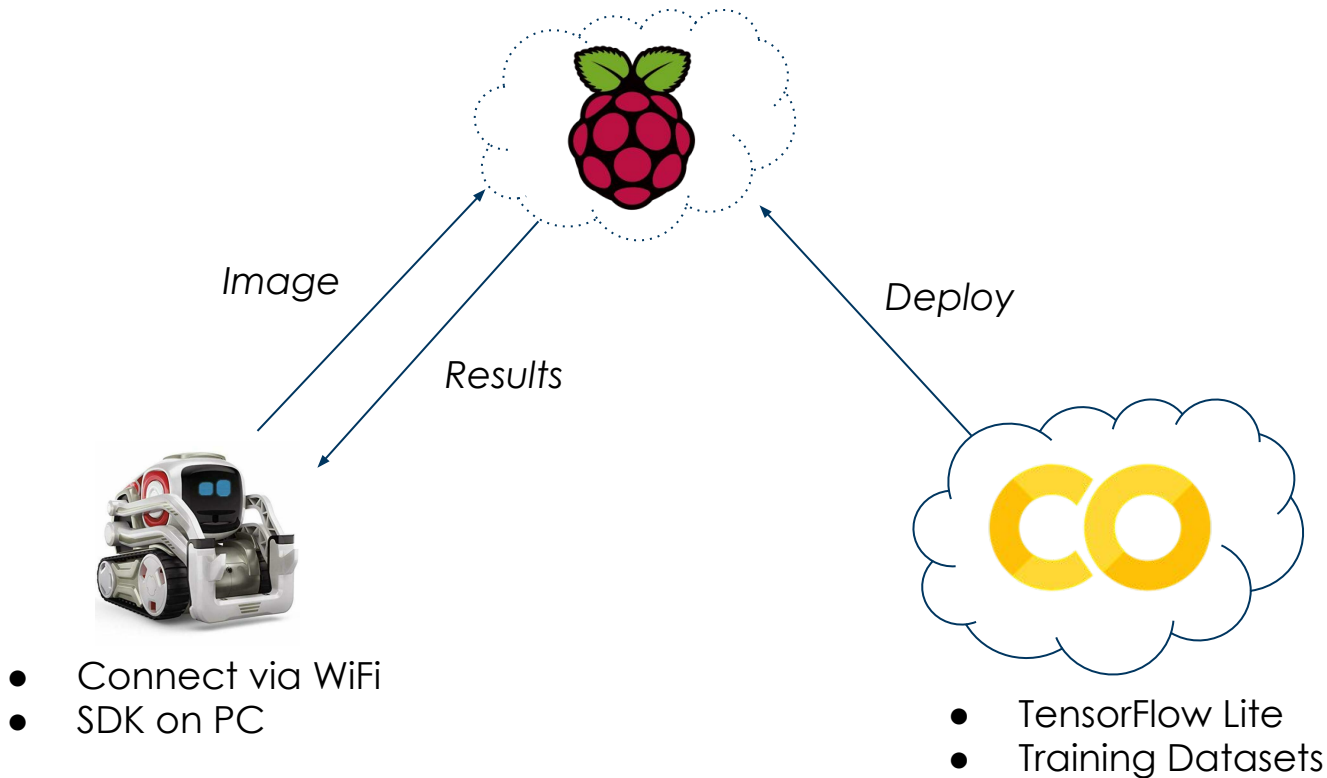
Edge



Cloud

Architecture

- Model & Labels
- TensorFlow Lite Interpreter



Cloud: Model Training and Data Storage

- **Google Colaboratory**

- Hosted Jupyter notebook service
- Free access to computing resources (GPUs)



- **TensorFlow Lite**

- Set of tools to help developers run TensorFlow models on mobile, embedded, and IoT devices.
 - TF Lite Converter
 - Converts TensorFlow models into an efficient form for use by the interpreter.
 - TF Lite Interpreter
 - Runs specially optimized models on mobile phones, embedded Linux devices and microcontrollers.

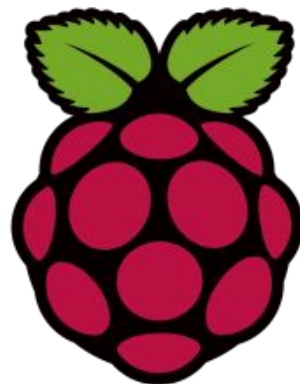
Edge: Handle Requests and Run Inference

- **Files**

- TensorFlow Lite Model & Labels
 - 13 MB
- TensorFlow Lite Interpreter
 - 5 MB
- IoT Image
 - 8 kB

- **Client Requests**

- Classify
- Exit



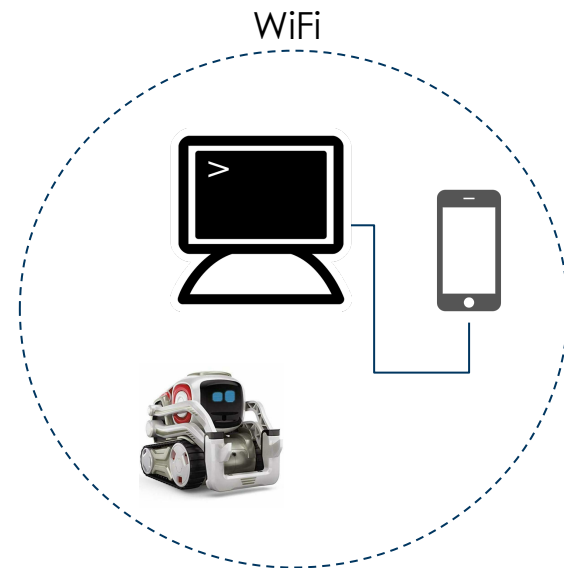
IoT: Cozmo Robot

- **System**

- Cozmo's engine packaged in iPhone app
- Cozmo connected to iPhone via WiFi network
- iPhone connected to PC via USB
 - Cozmo SDK ran on PC

- **Functions**

- Remote Control & Camera Feed
 - Controlled via Flask web page
- Send images
 - Send via scp to Edge



Classification: Off-the-shelf

- Pre-trained model and public dataset
- **Model:** Mobilenet
 - 8.9 MB
 - ~300 ms inference time
- **Dataset:** ImageNet 2012
 - 999 classes
 - Cup, Toaster, Stove, Microwave, Candle, etc.
 - ~1,000 training images/class



Classification: Rock, Paper, Scissors

- **Dataset:** Images from Cozmo
 - ~25,000 training images
 - ~4,000 raw images
 - ~21,000 augmented images
 - Random contrast/brightness/crop/resize/etc.



Paper



Paper



Rock



Scissors



Paper



Paper



Paper



Scissors



Scissors

Raw



Rock



Paper



Rock



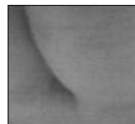
Paper



Paper



Scissors



Paper



Scissors



Paper

Resize



Rock



Paper



Scissors



Paper



Scissors



Paper



Scissors



Paper

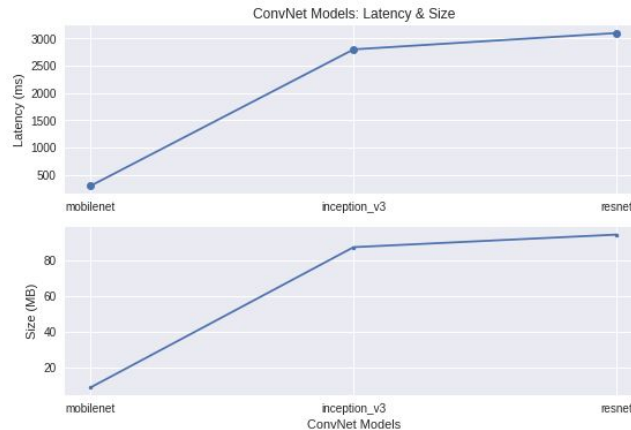
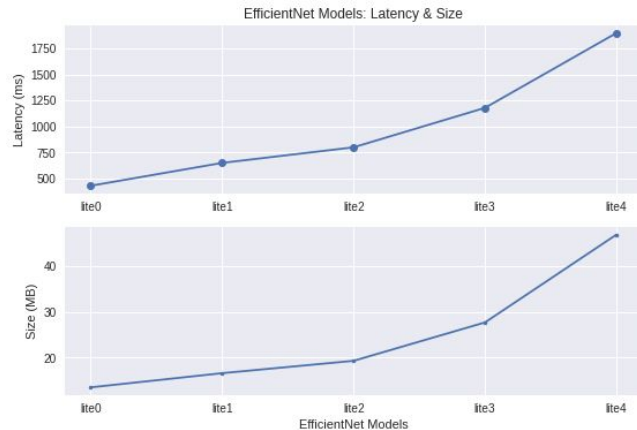


Scissors

Contrast

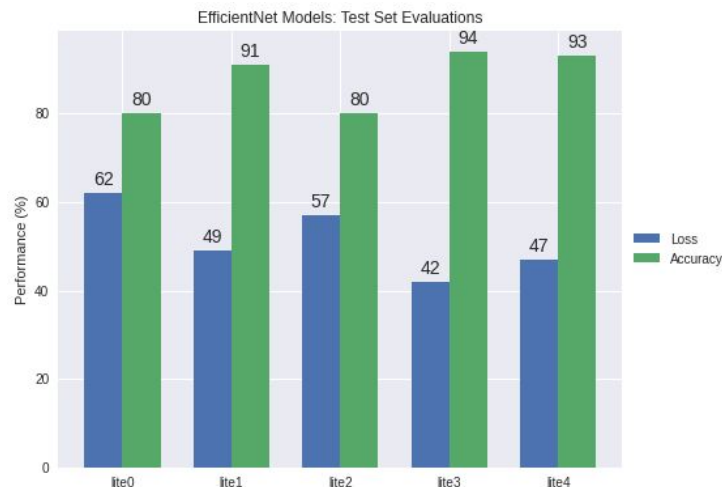
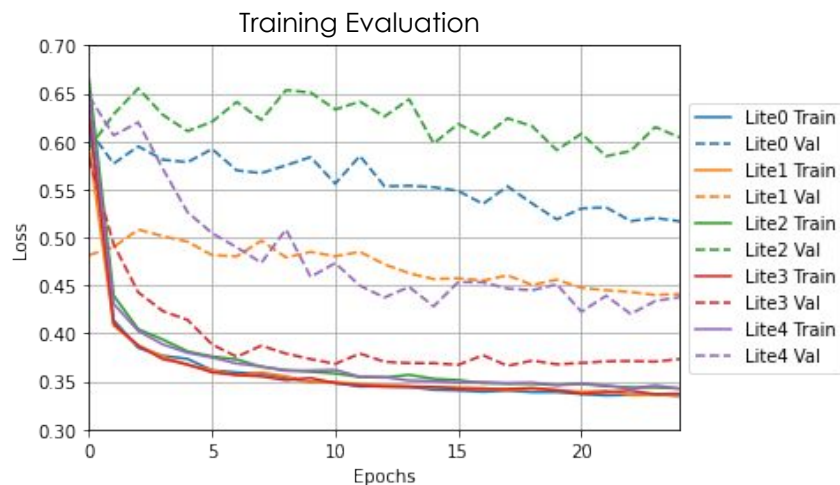
Classification: Rock, Paper, Scissors

- **Model:** EfficientNets
 - Mobile-size models
 - Quick inference
 - Minimize parameters and FLOPS, maximize accuracy
 - *Compound Scaling Method*



Classification: Rock, Paper, Scissors

- **Model:** EfficientNets
 - Fastest Inference: EfficientNet-lite0 (14 MB)
 - Highest Accuracy: EfficientNet-lite3 (28 MB)



DEMO